North American Journal of Fisheries Management 21:156–169, 2001

Restoration of Lake Trout in Parry Sound, Lake Huron

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Abstract.—The purpose of this study was to explain how a series of management actions facilitated the restoration of lake trout *Salvelinus namaycush* in Parry Sound of Lake Huron. Since 1988, the relative abundance of wild spawning lake trout has increased in Parry Sound. Estimates of the spawning population (1994–1997) varied between 16,000 and 29,000 individuals. This increase in abundance was achieved by controlling sea lampreys *Petromyzon marinus* (1960), stocking yearling lake trout derived from the remnant stock (1981–1997), creating a refuge (1987), restricting harvest by reducing both angling seasons and possession limits (1981), and imposing a size limit for retention (1994). By 1997, the three criteria of successful restoration were achieved, that is, the average age of mature females was 1 year older than the age of first maturation, wild fish composed 50% of the spawners, and the abundance of wild lake trout had been stable or increased for three consecutive years. As a consequence, stocking of yearlings was discontinued after 1997. A stochastic simulation model was developed to assess the likelihood of realizing sustainable harvests over a range of total mortality rates. We concluded that lake trout restoration is possible if sea lampreys are controlled, the appropriate strain is stocked, and exploitation is restricted, although additional limiting factors could hinder rehabilitation success in some locations of the Great Lakes.

Lake trout *Salvelinus namaycush* were either extirpated or greatly reduced in abundance in the Great Lakes because of the combined effects of exploitation and parasitism by sea lampreys *Petromyzon marinus* (Walters et al. 1980). The chronology of declines in lake trout abundance varied from lake to lake, but by the early 1950s, remnant stocks in Lakes Superior and Huron were the only stocks that remained (Hansen et al. 1995; Johnson and VanAmberg 1995). Following the introduction of hatchery-reared yearling lake trout and the application of effective control of sea lampreys, natural recruitment of lake trout was observed in Lakes Superior (Dryer and King 1968), Michigan (Jude et al. 1981), Huron (Nester and Poe 1984; Anderson and Collins 1995), and Ontario (Marsden et al. 1988). However, only in Lake Superior has substantial natural recruitment from wild and stocked spawning populations been documented (Krueger et al. 1986; Hansen et al. 1995).

Abiotic and biotic constraints may explain the dearth of natural recruitment in spawning populations of lake trout derived from hatchery stockings. Degraded spawning habitat (Casselman 1995; Edsall and Kennedy 1995) can explain the poor survival of lake trout eggs at some locations. Early mortality syndrome can suppress recruitment if the main prey is alewives *Alosa pseudoharengus* (Fitzsimons and Brown 1998). Predation of early life stages of lake trout (Jones et al. 1995; Krueger et al. 1995), sea lamprey predation on potential spawners (Johnson and VanAmberg 1995), excessive exploitation (Berst and Spangler 1972), phenotypic variation of lake trout (Perkins et al. 1995), and selection of poor quality spawning habitat by naïve spawners (Marsden et al. 1988) may also constrain natural recruitment. The relative importance of these abiotic and biotic constraints may vary both between and within the Great Lakes.

In Lake Huron, only the Parry Sound and Iroquois Bay stocks of lake trout survived after sea lampreys became established in the 1930s (Berst and Spangler 1973). Angling harvests of lake trout declined in Parry Sound during the late 1950s, reaching low levels by 1961. Mortality from sea lamprey parasitism was high, as evidenced by the high marking rates. Lake trout angling continued on a much reduced level through the 1960s but declined further in the 1970s. Very few spawning lake trout were observed on the spawning shoals from 1972 to 1978. The purpose of this study was (1) to document the management tactics used to restore the wild population of lake trout in Parry Sound and (2) to show how the recovery of a naturally recruited spawning population of indigenous lake trout was associated with these management actions.

Methods

Study Area

Parry Sound (9,207 ha) is the deepest (112 m maximum depth) bay along the eastern shore of

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Received September 21, 1999; accepted August 4, 2000

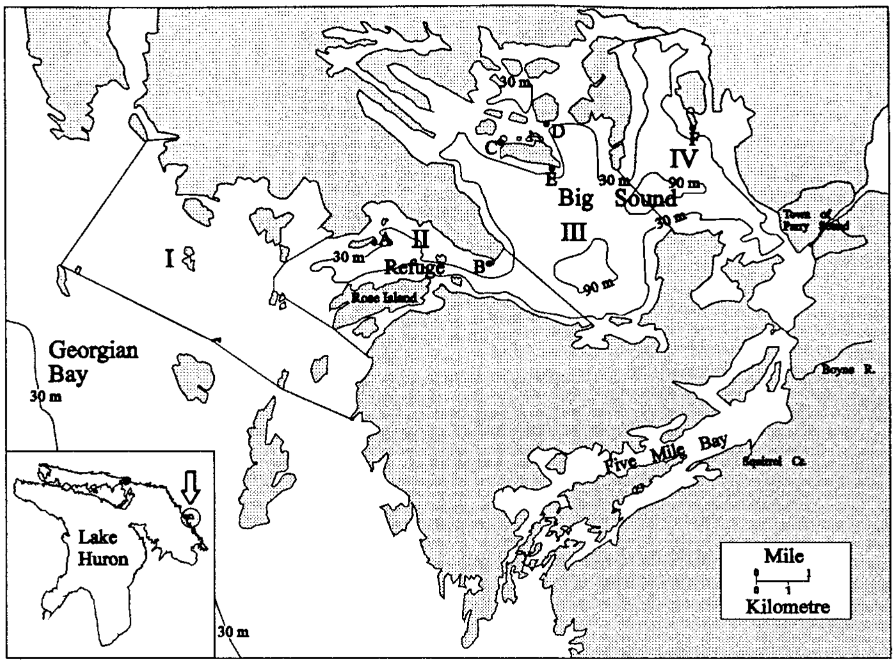

FIGURE 1.—Map of the Parry Sound area of Lake Huron showing fall trap-net locations and areas I–IV. Trap-net sites are as follows: A = Davy Island, B = Killbear Point, C = Bar Island, D = Horse Island, E = Mowat Island, F = Mary Island.

Georgian Bay (Figure 1). Much of the area exceeds depths of 90 m, and the average depth is 41 m. Parry Sound is connected to Georgian Bay by a shallower channel adjacent to Rose Island. For the purposes of this paper, we divided Parry Sound into two areas: the eastern portion (areas III and IV in Figure 1), which is locally referred to as the Big Sound; and the western portion (area II in Figure 1), hereafter referred to as the refuge. Parry Sound water is slightly basic (pH = 8.0–8.5), alkaline (34.2–54.7 mg/L), unproductive (total dissolved solids = 6.9 mg/L), relatively transparent (Secchi disk depth = 5.8 m), and oxygen saturated (10.8 mg/L at 109 m; Ontario Ministry of Natural Resources [OMNR], unpublished data, 1978).

Management Actions

Sea lamprey control and assessment.—Sea lamprey control in Lake Huron was initiated in 1960 (Smith and Tibbles 1980). The Boyne River was treated every 5 years starting in 1960, and Squirrel Creek was treated in 1969, 1971, 1972, and 1974 (Johnson 1988). Sea lamprey marking data were summarized by 10-cm length intervals for all lake trout captured (approximately 7,000 fish) in the fall trap nets from 1988 to 1998. Historical sea lamprey marking data were obtained from wild lake trout (73 fish) captured in small-mesh gill-net surveys conducted on the spawning shoals in 1958. Fresh (A1–A3) sea lamprey wounds were combined and reported as number of sea lamprey marks per 100 fish (Eshenroder and Koonce 1984). Sea lamprey-induced mortality (Sitar et al. 1999) was estimated as

$$M_{\mathrm{L}i,t} = W_{i,t+1} \cdot [(1 - P_i)/P_i], \qquad (1)$$

where $M_{\mathrm{L}i,t}$ is sea-lamprey-induced mortality for length-class i in year t, $W_{i,t+1}$ is the mean number of A1–A3 wounds per fish for length-class i sampled in year $t + 1$, and P_i is the probability of surviving a single sea lamprey attack for length-class i lake trout.

Stockings.—Splake lake trout *S. namaycush* × brook trout *S. fontinalis* were stocked in 1972, 1975, and 1976. Starting in 1981, yearling lake trout derived from wild Parry Sound lake trout and backcrosses (splake × lake trout) were stocked in Parry Sound; backcross stockings were discontinued in 1991, and lake trout stockings were stopped in 1997. Rainbow trout *Oncorhynchus mykiss* were stocked from 1986 to 1994 in Parry Sound to provide alternative fishing opportunities. No lake trout were stocked from 1989 to 1991 to control the spread of epizootic epitheliotropic virus disease (EEDV; Bradley et al. 1989; Stark 1996). Of all the lake trout and backcrosses stocked into Parry Sound, 63% were stocked close to the town of Parry Sound (in area IV), and areas II and III received 9% and 28%, respectively. Stocked lake trout and backcrosses were fin-clipped; the absence of clips was used to distinguish wild lake trout in all assessment catches.

Refuge for lake trout.—In 1987, a 1,061-ha refuge was created to exclude lake trout angling from what was then considered the core of the wild lake trout range. This refuge was expanded to 1,980 ha in 1990 (Figure 1) to increase the level of protection for wild lake trout.

Fishery restrictions and harvest assessment.—Daily harvest limits, length restrictions, fishing seasons, and area closures have been used to reduce angling mortality. The first closed season (October 1–November 30) for lake trout in Georgian Bay (including Parry Sound) was imposed in 1981. From 1982 to 1987 harvest of lake trout was permitted in Parry Sound between May 1 and September 30; winter fishing for lake trout was allowed only west of Killbear Point (site B in Figure 1). A winter fishery reinstated in the eastern portion of Parry Sound in 1987 (area IV) was expanded in 1997 (areas III, IV). Harvest seasons for lake trout are now February 8–March 31 and June 24–August 31, the fishery is closed from October 1 to December 31, and catch and release is allowed for the balance of the year. Daily catch and possession limits were reduced from three to two in 1987, and then to one trout not exceeding 61 cm total length in 1994. In 1997, the area protected by these regulations was expanded by 3,902 ha into Georgian Bay (area I), and the number of fishing lines allowed was reduced from two to one per angler during the winter season.

From 1982 to 1997, six summer roving creel surveys stratified by day, area, and period of day (Orsatti et al. 1991) were conducted on the waters inside and adjacent to Parry Sound to estimate the number of fish caught, harvested, and released alive. Nine roving stratified winter creel surveys of similar design were conducted on Parry Sound from 1985 to 1997. Total yield estimates (kg/ha) were based on an area of 8,303 ha, the area of lake trout habitat in the refuge and Big Sound.

Spawning Stock Assessment

From approximately mid-October (the onset of 12°C surface water temperature) to mid-November of 1988–1998, trap nets located close to spawning shoals (four sites in the Big Sound and two in the refuge; see Figure 1) were set to index spawner abundance, tag individual lake trout, and record morphological attributes. Trap nets (2.5 m^2, black nylon mesh, 51-mm stretch mesh) with 15-m to 25-m leads were set at depths of 2.5–3.0 m. Nets were generally lifted every other day, but depending on weather and fish abundance, set duration varied from 1 to 5 nights. Our index of relative abundance of lake trout was the geometric mean catch per net night. Length (mm), weight (g), sex, maturity, sea lamprey scars (A4) and wounds (A1–A3), fin-clip code, and tag information were recorded for each fish. Scales and pectoral fin rays were collected for age determination from 1996 from lake trout captured in trap nets at Davy and Horse islands. From 1994, lake trout were tagged with disk or anchor tags.

We used the adjusted Petersen method (Chapman 1951; Ricker 1975) to estimate the size of 1994–1997 spawning stock populations. Estimates were based on the number of lake trout tagged in fall trap nets (1994–1997) and recovered in trap nets (1995–1998) or gill nets (1998) in the following year. We assumed a tag loss of 20% based on our observations of double-marked lake trout recaptures in 1998 fall trap nets (unpublished data). Fall trap-net catches in the year following tagging were adjusted to exclude first-time spawners. Excluded fish were younger than age 8, and the number excluded by age was based on the maturation rate observed in the 1997 gill-net study. Movement of recaptured fish between tagging sites supports the assumption of random mixing (MacLean et al. 1981).

Population Assessment

Multimesh monofilament gill nets were set overnight in the hypolimnion in September 1997 (14 sets) to provide information on the age, sex, maturity, and growth of backcross, stocked, and wild lake trout. The gangs were 345 m in length and consisted of a 25-m panel of 38-mm stretched mesh and 46-m panels of stretch mesh sizes of 51, 64, 76, 89, 102, 114, and 127 mm. Six sites in the refuge and eight sites in Big Sound were selected randomly from areas having temperatures less than 12°C. Length (mm), weight (g), sex and maturity were recorded. Fecundity was estimated by determining the weight of two subsamples of 100 eggs and then multiplying the average number of eggs per gram of ovary by the whole ovary weight. The parameters of the relation between fecundity and body weight were used in the simulation model.

TABLE 1.—Parameter names, values, and descriptions for equations provided in the methods.

Parameter	Value	Description
Nat	0.12	Natural mortality
es_1	−0.1929	Slope of egg-to-yearling-survival equation
es_2	−3.8041	Intercept of egg-to-yearling-survival equation
L_v	10^8 m^3	Volume of Parry Sound
p_1	4.0	Maturity function shape parameter
p_2	2,000 g	Weight at 50% maturity
p_3	1.0	Asymptotic maturity
s_1	1.38	Selectivity shape parameter
s_2	−0.45	Selectivity shape parameter
s_3	0.01	Selectivity shape parameter

Otoliths and pectoral fin rays were taken from all lake trout for comparison of estimates of age.

Gill nets were set for short durations (1.5 h) in the nearshore zone in the spring of 1997 (spring littoral index netting, or SLIN) and 1998 (90 sets each year) and the fall of 1998 (fall littoral index netting, or FLIN; 72 sets). Each gang consisted of 91.7 m of either 38-, 51-, or 64-mm (stretched-measure) monofilament gill net 2.4 m deep with six panels per gang. The nets were set perpendicular to the shore from a depth of 2.5 m to a maximum of 50 m, depending on the morphology of the littoral zone. Nets were set at intervals throughout the day from ice-out until surface temperatures reached 13°C. All sites were selected at random within five spatial strata, four in the Big Sound and one in the refuge. Before release, all fish caught were weighed (g), measured (total and fork length; mm), examined for the presence of finclips, and tagged, and a pectoral fin ray was removed for age determination. The fall survey was conducted using the same protocol as the spring survey, beginning when the surface temperatures had cooled to 13°C.

Population Model

Incorporating the assumption that no additional lake trout would be stocked, a stochastic (lognormal distribution of variation) age-structured model (Table 1), based on estimates of the total abundance of stocked and wild lake trout in 1997, was used to predict the future abundance and age structure of the population:

$$N_{s,t+1,\ a+1} = N_{s,t,a} \cdot e^{-(\mathrm{Nat}+L_{t,a}+F_{t,a})}, \qquad (2)$$

where N is the number of stocked or wild lake trout (s), by year (t), and age (a); Nat is natural mortality, $L_{t,a}$ is sea lamprey mortality, and $F_{t,a}$ is fishing mortality.

The range of the stochastic distribution was based on variations (±25% of mean) of egg, larval, and adult natural survival (Walters et al. 1980). The range of egg survival was based on the survival of eggs experimentally deposited in spawning substrates over winter (B. Henderson, unpublished results). Annual variations of natural mortality were constrained to ±25% of the mean in this analysis, based on the ranges observed from inland lakes (Payne et al. 1990). Equilibrium conditions without harvest were used to iteratively estimate the parameters of the stock and recruitment function. A stock–recruitment relationship was simulated by making survival from the egg to yearling stage inversely related to the density of eggs. The main purpose of the stock–recruitment function was to keep maximum adult abundance within the estimated carrying capacity. Initially, the combination of parameters (equation 3) was determined that iteratively produced a stable biomass of lake trout approximately equal to the estimated carrying capacity (kg/ha) without exploitation provided from published estimates (Payne et al. 1990). Thus, we ran the model until we found a combination of slopes and intercepts that resulted in stable population sizes close to the estimated carrying capacity. This stock–recruitment equation (equivalent in function to a Ricker stock–recruitment equation) was used in all subsequent simulations, using varying levels of harvest. Egg-to-yearling survival varied from 0.002 at low egg density to 0.0007 at high egg density. The equation is

$$\mathrm{ES}_t = \mathrm{es}_1 \cdot \log_e(\mathrm{teggs}_t/L_v) + \mathrm{es}_2, \qquad (3)$$

where ES_t is egg survival, teggs_t is total eggs, L_v is volume of water in Parry Sound, es_1 is the slope of the regression, and es_2 the intercept (Table 1). The ranges of egg-to-yearling survival rates are similar to those of Matuszek et al. (1990). The proportion of mature females, $\mathrm{Pmat}_{t,a}$, of age a during year t was estimated by

$$\mathrm{Pmat}_{t,a} = \frac{p_1 \cdot W_{t,a}^{p_2}}{p_3 + W_{t,a}^{p_2}}, \qquad (4)$$

where p_1 to p_3 were shape parameters (Table 1), and $W_{t,a}$ was weight. Egg production estimates were a function of the proportion of mature females by weight class and fecundity. The lowest weight at maturation was 2 kg. Thus, total egg production (teggs_t) was estimated as

TABLE 2.—Estimates of sea-lamprey-induced instantaneous mortality rates in Parry Sound and the main basin of Lake Huron (sample size in parentheses).

Size group (total length, mm)	Parry Sound		Main basin, Michigan waters		
	1958	1988–1998	North[a] 1991–1998	North central[b] 1991–1998	South central[b] 1991–1998
<432		0 (1)	0 (285)		
432–532		0 (19)	0.149 (152)	0 (3)	
533–634	0.403 (3)	0.001 (1,538)	0.281 (308)	0.134 (149)	0.159 (93)
635–734	0.319 (26)	0.002 (2,381)	0.303 (583)	0.229 (145)	0.155 (278)
>734	0.581 (44)	0.006 (629)	0.262 (179)	0.213 (27)	0.335 (277)

[a] Data provided by Mark Ebener, Chippewa/Ottawa Treaty Fishery Management Authority.
[b] Data provided by Aaron Woldt, Michigan Department of Natural Resources.

$$\text{teggs}_t = \sum_{a=\alpha}^{a=\omega} N_{t,a} \cdot W_{t,a} \cdot \text{sr} \cdot \text{Pmat}_{t,a} \cdot E_{t,a}, \qquad (5)$$

where $N_{t,a}$ is the number of lake trout by year t and age a (where α = age of first maturation and ω = age of oldest female), $W_{t,a}$ is weight, sr is sex ratio (0.5), and $E_{t,a}$ is fecundity by weight of females.

Mortality from sea lamprey parasitism was not modeled explicitly (i.e., $L_{t,a} = 0$ for all combinations of t and a) because it was a small fraction of the total mortality (Table 2) and the stochastic variation of natural mortality would exceed the anticipated levels of sea-lamprey-induced mortality. Angling harvest rates were varied from 0.0 to 0.5 kg/ha to determine the probabilities of sustainable harvests. The existing angling harvest in 1997 was 0.37 kg/ha, but harvests ranging from 0.0 to 0.5 kg/ha were considered. The results of harvesting at varying levels were calculated in the model as instantaneous fishing mortality rates. Harvests were restricted to lake trout in the range of 20–61 cm, but the effects of catching and releasing trout greater than 61 cm were simulated with a hooking mortality of 0.15 (Loftus et al. 1988; Dextrase and Ball 1991). Within the range of 20–61 cm a generalized gamma function (Henderson and Wong 1991) was used to simulate size selectivity by anglers:

$$S_a = \frac{s_1 \cdot e^{s_2 W_a} \cdot W_a}{1 + s_1 \cdot s_3 \cdot e^{s_2 W_a} \cdot W_a}, \qquad (6)$$

where S_a was selectivity at age a, s_1–s_3 were shape parameters (Table 1), and W_a was weight at age so that $\Sigma S_a = 1.0$. The selectivity by age and weight-class was used to allocate harvests and estimate fishing mortalities by age-class. Log-normally distributed natural mortality varied stochastically around an average of 0.12. Growth, maturity, and fecundity were based on the samples of Parry Sound lake trout collected by gill nets in September 1997. Increments of growth varied stochastically ($\pm$25%); yearly increments were determined from a Walford equation, with an asymptotic weight of 7 kg. Probabilities of extinction were calculated over a range of instantaneous fishing mortality rates. Extinction was operationally defined as occurring when the average female spawning stock abundance fell below 100 individuals between years 50 and 75 of each 100-year simulation. It took 50–75 years to predict the consequences of fishing mortality. Sustainability of exploitation was assessed by comparing the probabilities of extinction with fishing mortality rates and harvest rates over 100 years and by repeating the simulations 100 times for each level of harvest.

Results

Management Actions

Sea lamprey mortality.—In 1958, before the major decline in the fishery in Parry Sound, sea lamprey parasitism was 45 times higher than it was from 1988 to 1998. For wild lake trout (533–734 mm total length, N = 29) sampled in October 1958, marking rates varied from 33 to 39 marks (A1–A3) per 100 fish. In contrast, the average marking rates in the same size-class (N = 3,919) did not exceed 1 sea lamprey mark per 100 wild lake trout sampled during October 1988–1998. Sea lamprey-induced mortality estimated from the 1958 sample of lake trout larger than 533 mm varied from 0.319 to 0.581 (Table 2). Instantaneous mortality caused by sea lamprey parasitism estimated from the lake trout sampled in Parry Sound (1988–1998) was extremely low (0.001–0.006).

Fishery restrictions and harvest.—Stocked backcrosses and lake trout (Table 3) were first harvested by anglers in 1985 (Table 4) and by 1988 had recruited to the spawning stock (Figure 2). Although initially more backcrosses than lake

TABLE 3.—Number of yearlings, by species or hybrid, stocked into Parry Sound from 1981 to 1997.

Year	Lake trout	Backcross	Rainbow trout
1981	24,100	95,646	
1982	10,600	97,000	
1983		60,000	
1984	33,800	72,500	
1985	19,100	61,400	
1986	33,500	48,300	3,000
1987	30,100	39,700	3,000
1988	31,800	48,200	5,000
1989		20,000	18,500
1990		15,000	5,000
1991		20,550	4,000
1992	12,850		
1993	43,200		
1994			
1995	27,888		
1996	46,251		
1997	60,258		

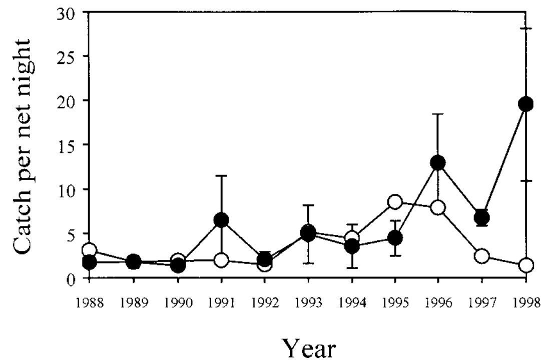

FIGURE 2.—The fall trap-net catch per unit of effort of wild (solid circles) and stocked (clear circles) lake trout in Parry Sound during the period October 15–31 in 1988–1998. Bars indicate standard errors.

trout were caught, the catch per number stocked was 5 times greater for lake trout. Backcross stocking was terminated in 1991 due to poor returns and concerns about mixed-strain stocking (Evans and Willox 1991; Jones et al. 1995). There were no reported angler catches of backcrosses after 1994 (Table 4). Annual angling effort (610–37,000 rod-hours) and harvest (0–403 wild lake trout) increased from 1982 to 1985.

Although the closure of the Big Sound winter fishery from 1982 to 1987 reduced the winter harvest, the combined summer and winter yield was still high (Table 4). With the opening of a portion of the Big Sound to winter fishing in 1987, winter fishing effort (2,587–18,025 rod hours) and harvest (336–2,128 lake trout and backcrosses) increased. Stocked lake trout and backcrosses accounted for most of the increased harvests, presumably because wild lake trout were protected in the refuge. The annual yield was reduced to 0.76 kg/ha by 1988. Expansion of the refuge area in 1990 provided further protection to the native stock and annual yield was reduced to 0.54 kg/ha by 1993.

Regulation changes in 1994 had an immediate effect on the lake trout fishery. Summer fishing effort was reduced by one-half and harvest by two-

TABLE 4.—Estimated effort (angler hours for backcross and lake trout), harvest, percent released, total killed (including hooking mortality), and total yield of wild and stocked lake trout in Parry Sound during the summer and winter from 1982 to 1997.

Season and year	Effort (angler hours)	Number harvested: Wild fish	Stocked fish	Backcross	Percent released	Total killed	Yield (kg/ha)
Summer							
1982	610	0	0	0	0	0	0
1985	33,937	323	460	2,386	23	3,304	1.01
1988	22,429	51	775	806	29	1,722	0.34
1993	9,680	267	533	133	25	976	0.25
1994	5,363	70	47	70	87	369	0.08
1997	7,712	642	286	0	70	1,228	0.21
Winter							
1985	2,587	80	0	256	13	344	0.06
1987	18,025	94	579	1,455	16	2,187	0.40
1988	24,212	107	993	1,092	23	2,289	0.42
1991	30,094	57	980	386	21	1,479	0.23
1992	42,212	122	1,785	492	28	2,540	0.43
1993	36,832	140	1,125	204	34	1,581	0.29
1995	11,553	207	133	0	60	417	0.07
1996	11,496	394	110	0	49	577	0.08
1997	10,561	529	340	0	57	1,043	0.16

TABLE 5.—Numbers of sexually mature lake trout (wild and stocked) in Parry Sound, October–November 1994–1997 determined from Petersen estimates and 95% confidence limits (CL).

Year	Estimated number	95% CL	Number	
			Tagged	Recaptured
1994	16,698	9,680–31,309	604	11
1995	17,050	11,875–25,393	806	27
1996	20,423	14,523–29,706	1,400	31
1997	29,916	17,342–56,092	1,037	11

thirds. Winter fishing effort and harvest were reduced by two-thirds (Table 4). The percentage of lake trout released alive increased from less than 34% to more than 50% after 1994. The annual lake trout yield was reduced to 0.37 kg/ha by 1997.

Spawning-Stock Assessment

The average catch of stocked lake trout per trap-net night (CUE) increased to a peak in 1995 and then declined (Figure 2). Wild lake trout always predominated in the catches from Davy Island. By 1997 the proportions of wild lake trout at the other five sites in the Big Sound exceeded 50%. On average, the CUE of wild lake trout increased from 1988 to 1998, particularly after 1995. Petersen estimates of the spawning population of lake trout increased from 16,700 fish in 1994 to 29,900 fish in 1997 (Table 5).

In 1996–1998, the wild spawning stock consisted of at least eight year-classes (Table 6), modal ages (fin rays) being 7–8 for females. Ages determined from otoliths and fin rays were significantly correlated (otolith age = 0.625 · fin ray age + 0.953, df = 179, r^2 = 0.71, slope $P < 0.0001$, intercept $P = 0.01$). A paired *t*-test showed a significant difference between fin ray (age 5.6) and otolith (age 5.2) ages (mean difference = 0.37 years, $t = 5.292$, $P < 0.001$). Fin ray age underestimated otolith age by less that 1 year. No evidence of a divergence in age assessment from the two structures was detected among older lake trout.

Population Assessment

The density (CUE = catch per set) of wild lake trout was higher in the refuge (17.0 ± 3.46 [SE], $N = 6$ sets) than in the Big Sound (8.14 ± 1.317, $N = 8$ sets), as determined from deepwater gill-net assessment in 1997. The proportion of wild lake trout exceeded 0.8 in both areas and was highest in the refuge. Nine year-classes of wild lake trout were captured (modal age = 6; Figure 3). Wild lake trout increased in fork length from 200 mm at age 2 to 600 mm by age 9 (Figure 4; $\log_e$ weight = −11.736 + 3.082 · $\log_e$ fork length, df = 157, r^2 = 0.97, $P < 0.001$). Females mature (50%) at age 7 and males at 6 years (Figure 5). Fecundity (total number of eggs) varied from 1,500 to 4,500 for females weighing 2–3.2 kg (fecundity = −10,093 + 23.64 · fork length, df = 18, r^2 = 0.23, $P = 0.02$).

The spring abundance (SLIN) of wild lake trout (geometric mean of catch per set, GEOCUE) increased from 1997 to 1998 in Big Sound but not

TABLE 6.—Age frequencies (determined from fin ray ages) of Parry Sound lake trout captured in trap nets set during the spawning season.

Year	Origin and sex	Number of fish by age											
		4	5	6	7	8	9	10	11	12	13	14	15
1996	Stocked												
	Male	4	9	2	2	9	28	23	4	4		1	
	Female				3	10	17	10	3		2		
	Wild												
	Male	6	12	111	184	83	17	1	2	3	1		
	Female		1	2	47	41	18	5	4	5			
1997	Stocked												
	Male		1	2	9	6	6	6	2				
	Female				2	10	8	1	5				1
	Wild												
	Male		12	50	120	52	13	3	2				
	Female			7	36	28	8	5	1	1			
1998	Stocked												
	Male	1	3	2	3	3	7	4	3	1			
	Female				1	4	6	2	2				
	Wild												
	Male		2	35	124	122	46	10	5	4	2		
	Female				20	54	30	5	2	1		1	

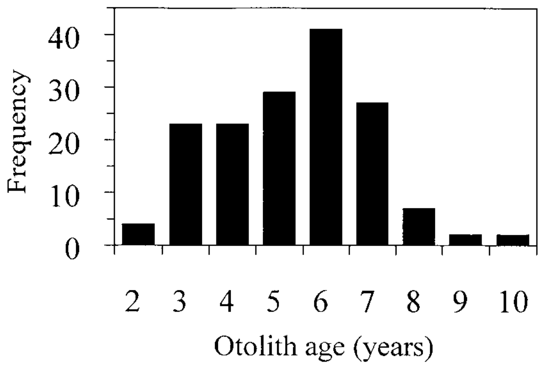

FIGURE 3.—Age composition of Parry Sound wild lake trout captured in multimesh monofilament gill nets set in September 1997.

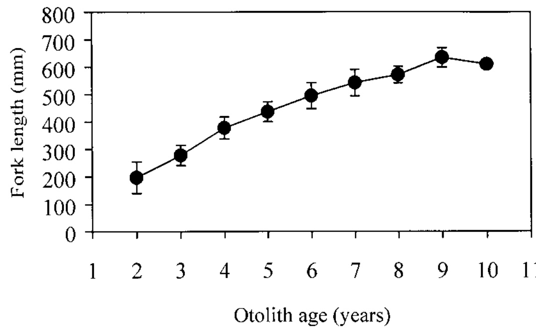

FIGURE 4.—Fork length and standard errors by age of Parry Sound wild lake trout captured in multimesh monofilament gill nets set in September 1997.

in the refuge (Table 7). The spring density of both wild and stocked lake trout was higher in Big Sound than in the refuge. In the fall (FLIN), the refuge had twice the density of wild lake trout as the Big Sound but fewer stocked lake trout.

Population Model

The probability of extinction increased exponentially with fishing mortality (Figure 6; fishing mortality rates are the average rates from ages 3–6). Average yields did not increase linearly with fishing mortality because the probability of extinction increased with fishing mortality. The existing management expectation of harvests in the range of 0.3–0.35 kg/ha (total catch of 848 trout per year plus 400 expected deaths from hooking mortality) is probably sustainable, although the estimated chance of failure is 13%. This harvest can be achieved with an average fishing mortality rate of 0.20 for ages 3–6 ($Z = 0.32$, $M = 0.12$).

Discussion

Sea Lamprey Control

The extirpation of lake trout from most of Lake Huron by the mid-1950s (Berst and Spangler 1972) has been attributed to the combined effects of sea lamprey predation and commercial exploitation (Walters et al. 1980; Coble et al. 1990; Eshenroder 1992). Several factors appear to have delayed the effects of sea lamprey predation on Parry Sound lake trout. The invasion of sea lampreys into Georgian Bay was delayed compared to the main basin of Lake Huron (Henderson and Payne 1988). The relatively shallow and narrow passageways connecting Parry Sound to Georgian Bay slowed the immigration of sea lampreys from Georgian Bay. Finally, the closest sea lamprey spawning tributaries to Parry Sound are the Boyne River and Squirrel Creek in Five Mile Bay (Figure 1; Johnson 1988), restricting the potential abundance of locally produced sea lampreys. Although these factors delayed the effects of sea lampreys, the decline of Parry Sound lake trout in the 1960s was associated with high sea-lamprey-induced mortality.

Larger lake trout survive sea lamprey attacks better than smaller fish (Pycha and King 1975; Swink 1990). The absence of a commercial fishery in Parry Sound prevented the combined effects of sea lamprey parasitism on smaller lake trout and the exploitation of larger and older lake trout, factors that led to the extirpation of other stocks in Lake Huron (Coble et al. 1990; Eshenroder 1992). In the absence of commercial exploitation, larger Parry Sound lake trout survived until the 1960s, when sea lampreys were first controlled (Johnson 1988). The combined effects of an existing angling fishery and continued mortality by a residual sea lamprey population prevented any recovery of the

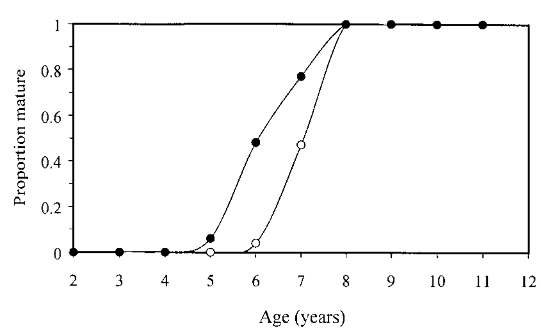

FIGURE 5.—Proportions of mature male (solid circles) and female (open circles) wild lake trout in Parry Sound captured in multimesh monofilament gill nets set in September 1997.

TABLE 7.—Relative abundance of wild and stocked lake trout in Parry Sound (Big Sound and Refuge) from short gill-net sets in the spring (SLIN) and fall (FLIN) of 1997 and 1998.

Year and season	Location	Stocked		Wild	
		LNCUE[a]	GEO-CUE[b]	LNCUE[a]	GEO-CUE[b]
1997					
SLIN	Big Sound	0.279 ± 0.0431 (72)	0.321	0.381 ± 0.0503 (72)	0.464
	Refuge	0.185 ± 0.0484 (18)	0.203	0.353 ± 0.1011 (18)	0.423
1998					
SLIN	Big Sound	0.390 ± 0.0732 (72)	0.476	0.488 ± 0.0722 (72)	0.629
	Refuge	0.252 ± 0.1285 (18)	0.286	0.339 ± 0.1242 (18)	0.403
FLIN	Big Sound	0.370 ± 0.0724 (54)	0.447	0.405 ± 0.0756 (54)	0.499
	Refuge	0.277 ± 0.0825 (18)	0.319	0.727 ± 0.1667 (18)	1.068

[a] Average of natural logarithm of catch per set + 1, plus or minus the standard error; number of sets in brackets.
[b] Geometric mean of catch per set.

lake trout during the 1960s and 1970s. Sea lamprey parasitism on spawning lake trout was negligible in Parry Sound from 1988 to 1998 compared with other areas of Lake Huron (Table 2). Thus, there has been no evidence of sea lamprey parasitism constraining lake trout rehabilitation in the Parry Sound area, at least not after 1988.

Hatchery Stocking

The role that hatchery stocking played in the rehabilitation of Parry Sound lake trout is difficult to assess. To distinguish contributions from stocked and native lake trout we must look at the timing of events. The first lake trout stockings should have reached sexual maturity in 1987. By 1988, stocked lake trout composed 59% of the lake trout catches in trap nets set for spawners, the majority of wild fish being observed at Davy Island. We suspect that stocked lake trout contributed to recruitment, possibly to the 1988 year-class and

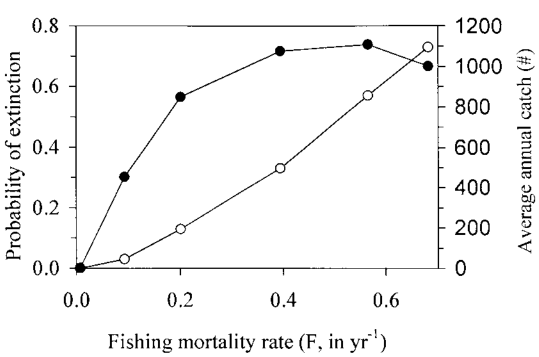

FIGURE 6.—Probability of extinction (open circles) and average annual catch in numbers of lake trout (solid circles) in relation to fishing mortality rate. Values were derived from a simulated population of lake trout in Parry Sound.

certainly to the 1989 and later year-classes. If recruitment of mature stocked lake trout to the spawning population did enhance natural recruitment, then increased abundance of wild spawners should have occurred by 1995. By the mid-1990s young wild lake trout were captured by anglers in Big Sound where they were not encountered previously. Increases in the proportion of wild lake trout were evident from the 1995 winter and 1997 summer creel surveys (Table 4) and from 1996 and 1997 fall trap-net catches in Big Sound (Figure 2). The timing and location of increased abundance of unmarked lake trout suggested that stocked lake trout did contribute significantly to natural recruitment. Studies conducted in Lake Superior also indicated that stocked lake trout contributed to natural reproduction, particularly in areas where significant nearshore spawning habitat was available (Hansen et al. 1997).

The recommended stocking rate for lake trout rehabilitation in Lake Huron is 2.5 yearlings/ha in water less than 73 m (Ebener 1998). In comparison, in Parry Sound from 1981 to 1997 a mean of 4.5 yearlings/ha were planted in the years stocking occurred. These high stocking rates may have initially accelerated restoration but could have increased intraspecific competition and, possibly, cannibalistic activity between young wild and stocked lake trout in later years when natural recruitment was significant (Evans and Willox 1991; Krueger and May 1991; Jones et al. 1995; Dunlop and Brady 1998). The absence of lake trout stocking from 1989 to 1991 corresponded to years of elevated natural recruitment. The 1989 and 1990 year-classes were well represented in the spawning population in 1996–1998 (Table 6); however, because females recruit at age 7 the predominance of age-7 and age-8 females in 1997 and 1998 is not unexpected. Subsequent assessment will better identify the relative strengths of year-classes totally derived from natural fish, but preliminary results suggest that recruitment of wild lake trout was stronger during the 3 years without stocking.

Harvest Restrictions

Restoration of the Parry Sound lake trout population by natural recruitment in the 1990s appears to be explained largely by management interventions to reduce mortality. The combination of area closures, establishment of a refuge, angling season reductions, harvest and possession limit restrictions, and sea lamprey control contributed to a reduction in total mortality in the spawning population and the subsequent increased recruitment of lake trout. The establishment of the refuge near Davy Island in 1987 was associated with an increased index of spawner abundance after 1990 (Figure 2). We suggest that this increase in abundance occurred because fishing mortality was reduced on the lake trout (ages 3–6) before their recruitment to the spawning population at age 7. The use of a refuge area in Lake Superior resulted in increased lake trout survival (Schram et al. 1995). Reductions in possession limits from 3 to 1, a maximum size limit of 61 cm, and shortened harvest seasons in the Big Sound area in 1994 were followed by increased abundance of spawners (Figure 2).

Many of the management actions employed in Parry Sound redirected angling to focus more on stocked than wild lake trout. The stocking of lake and rainbow trout to facilitate redirection of angling effort away from wild lake trout had the reverse effect; instead, it attracted more fishing effort, producing higher catches of wild lake trout (Table 4). Increased harvests of wild fish from effort directed at stocked fish have been observed elsewhere (Argue et al. 1983; Evans and Willox 1991).

Although the angling restrictions introduced in 1994 had the desired effect of reducing effort and harvest (Table 4), as fishing quality improved more anglers accepted the stringent regulations, and by 1997 annual harvest (0.37 kg/ha) exceeded the target level (0.33 kg/ha). Monitoring of future angling effort and associated lake trout mortality will be required. If necessary, a more direct method of restricting harvest may need to be considered, such as a finite tag system (annual issuance of a predetermined number of tags, each to be affixed to a harvested lake trout) that would reduce the potential of increased angling effort leading to increased mortality. The alternative would be the less appealing need for more restrictive regulations that indirectly and less efficiently limit angling mortality. The decision to discontinue stocking increases the need to control harvest on the lake trout population that is now dependent on natural reproduction.

Environmental Factors

Other factors may also have contributed to the recovery of lake trout in Parry Sound. Fitzsimons (1995) first demonstrated a relationship between thiamine deficiency and early life stage mortality of lake trout. Higher (6 times) levels of thiaminase were found in alewives than in rainbow smelt *Osmerus mordax.* These two prey species predominate the diets of lake trout in the Great Lakes, and as a consequence could lower thiamine levels in lake trout eggs, reducing fry survival (Honeyfield et al. 1998). Alewives are present in Big Sound but at lower densities (0.13 kg/ha) than rainbow smelt (33.42 kg/ha; G. L. Curtis, U.S. Geological Survey, personal communication). Early mortality syndrome should not significantly affect recruitment in Parry Sound because thiamine concentrations in lake trout eggs exceeded threshold levels (Fitzsimons and Brown 1998). Alewife predation on larval lake trout is probably not a constraint on recruitment (Jones et al. 1995; Krueger et al. 1995) because alewife abundance is relatively low.

Restoration

Three criteria are recommended for assessing the progress of lake trout restoration in the Great Lakes: (1) the average age of mature females should be 1 year greater than the average age at first maturity, (2) wild fish should compose at least 50% of the catch of spawning-size lake trout, and (3) wild fish abundance should be stable or increasing for three consecutive years (Hansen 1996; Ebener 1998). In Parry Sound all three criteria were met. The ages of wild female spawners ranged from 7 to 12 years in 1998 (modal age = 8), and 50% of the females were mature at age 7. By 1997, the proportion of wild lake trout reached or exceeded 50% at all six trap-net sites. The spawning stock has increased in abundance since 1994. Parry Sound lake trout were considered successfully rehabilitated by 1997 and stocking was terminated.

In comparison, other attempts at lake trout restoration in Lake Huron have been unsuccessful. Excessive angling harvests of lake trout have prevented rehabilitation in Iroquois Bay. Although minor angling regulation changes were made in Iroquois Bay, few wild fish were observed in the 1990s and exploitation is still too high for rehabilitation of the wild population (A. Liskauskas, OMNR, personal communication). In South Bay, high exploitation hindered development of a spawning population of stocked lake trout (Anderson and Collins 1995). Harvest controls in 1984 resulted in reduced mortality, increased adult abundance, and natural recruitment in 1990. However, exploitation has increased since 1990, and there is evidence of declining abundance of wild lake trout in South Bay (OMNR, unpublished data, 1997). Similarly, lake trout rehabilitation in proximity to the Bruce Peninsula was hindered by excessively high mortality rates (Mohr et al. 1997). Natural reproduction from stocked lake trout was observed near Rockport, Michigan, in 1982 (Nester and Poe 1984). Nearby a spawning population of predominantly unclipped lake trout was discovered on Mischley Reef in Thunder Bay in 1992 (Johnson and VanAmberg 1995). Rehabilitation of lake trout stocks in the Thunder Bay area is constrained mainly by mortality from sea lamprey parasitism, although excessive angling mortality is an additive cause and alewife predation at early life stages is also suspected (A. Woldt, Michigan Department of Natural Resources, personal communication). Wild lake trout fry were first observed in 1994 on Six Fathom Bank in central Lake Huron (Bowen et al. 1995). This recruitment was attributed to Seneca strain lake trout that are more resistant to sea lamprey predation. Recently, natural recruitment declined, perhaps a result of reductions in the stocking of Seneca lake trout (Johnson and Woldt 1999).

Management of a Restored Population

Sustainable yields of 0.5 kg/ha (Healey 1978) were expected initially for a restored population of lake trout in Parry Sound. A more conservative harvest of 0.33 kg/ha was chosen arbitrarily as a target to expedite the restoration of lake trout. A stochastic simulation model of the Parry Sound lake trout population was developed as a tool to assess the probability of sustaining various levels of harvest (Figure 6). We assumed that the instantaneous natural mortality rate (M) was 0.12 (Shuter et al. 1987). To corroborate this estimate, we estimated that $M = 0.14$ based on the 10 largest trout captured (L_∞ = 945 mm; 1981–1987) and an omega ($\omega = kL_\infty$) of 11.41 (Shuter et al. 1998). This estimate of M may be inflated slightly because k was estimated from a 1997 sample of the recovering Parry Sound lake trout population.

The ratio of fishing (F) to natural mortality (M) can be used to evaluate the risks of overexploitation. The anticipated harvest of 0.3–0.35 kg/ha in Parry Sound equates to a fishing mortality rate that is about 1.7 times the natural mortality rate. This ratio of F to M is higher than recommended (Deriso 1987; Walters and Maguire 1996); the ratio should lie in the range of 1.2–1.5, based on the ratio of M (0.12) to the von Bertalanffy growth coefficient (K = 0.196 and 0.129 for asymptotic sizes of 700-mm and 900-mm fork lengths, respectively) (Deriso 1987). The largest trout observed in 1997 was 700 mm, but the maximum size of lake trout sampled from a creel survey in 1965 was 900 mm. In comparison, the ratio of F to M for inland lake trout populations was closer to 1.2 (Payne et al. 1990). If a smaller F were used to achieve a lower ratio, then both the sustainable harvest and probability of extinction would be reduced. Sustainability of harvest in the range of 0.30 kg/ha was supported by modeling results. In comparison, harvests of 0.27 kg/ha in an inland lake trout population in Lake Opeongo were sustainable (Shuter et al. 1987; Matuszek et al. 1990). Although harvests of 0.30 kg/ha were considered achievable, lower harvests have lower risks. Currently, sea lamprey parasitism is low in Parry Sound, but increased sea-lamprey-induced mortality would necessitate a reduction in the fishing mortality rates.

In summary, the lake trout in Parry Sound are the only successfully restored stock in the lower four Great Lakes. Although the presence of a remnant lake trout stock appears to increase the likelihood of rehabilitation (Hansen et al. 1995; Krueger et al. 1996), successful natural reproduction has been achieved when the mortality of stocked populations was sufficiently controlled (Nester and Poe 1984; Anderson and Collins 1995; Bowen et al. 1995; Hansen et al. 1995; Johnson and VanAmberg 1995). The rehabilitation success at Parry Sound was accomplished by reducing sea-lamprey-induced mortality, stocking lake trout derived from the remnant wild stock, and sufficiently restricting harvests. Although additional limiting factors could hinder rehabilitation success in other locations, the Parry Sound experience provides an example of successful techniques that have achieved lake trout rehabilitation.

Acknowledgments

We thank the staff of the Parry Sound office of the OMNR, and in particular Lloyd Thurston and

Eric McIntyre, who were very active in the early stages of lake trout rehabilitation in Parry Sound and provided a substantial part of the early data used in this study. The assistance of Lake Huron Management Unit and Research staff during field and laboratory work is much appreciated. The Big Sound Lake Trout Public Advisory Committee is applauded for their dedication, support, and advice in the rehabilitation process. We thank Michael Hansen, Chuck Madenjian, Robert Payne, Nigel Lester, and three anonymous reviewers for useful comments on earlier drafts. This manuscript is Ontario Ministry of Natural Resources contribution 99-09.

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